Raj Reddy

(352) 530-3397 rajreddy23@outlook.com github.com/rjrddy linkedin.com/in/raj-reddy-1 raj-reddy.com

Education

University of Utah **Salt Lake City, UT**
B.S. Computer Science *August 2020 – May 2025*

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies **Dallas, TX**
Associate Software Engineer *June 2025 – Present*

- Developed real-time signal processing algorithms in Python and C++ on embedded systems, applying DSP techniques for defense sensor data streams.
- Built microservices for high-frequency data ingestion and storage, leveraging Podman and serverless infrastructure on AWS.
- Automated CI/CD pipelines with static code analysis and integration testing.

Doxy.me **Charleston, SC**
Software Engineer *August 2024 – May 2025*

- Built a real-time mental health inference system for Doxy.me's telehealth platform, fine-tuning Llama models to predict depression risk, anxiety, and mood indicators from tone, voice, and facial cues captured over live WebRTC sessions, served via SageMaker and Fargate.
- Improved prediction accuracy of the mental health inference system by applying RL feedback loops over structured clinical signals and physician-validated outcomes from live video consultations.
- Eliminated environment drift and enabled zero-downtime releases by standardizing CI/CD configuration with YAML-based pipelines across the ML stack.

University of Utah – VAAST Lab **Salt Lake City, UT**
Undergraduate Research Assistant *August 2024 – May 2025*

- Integrated photorealistic nature environments into head-mounted displays using Unity, building scene loading pipelines, asset configuration, and rendering setups for spatial perception experiments run with human participants.
- Implemented continuous 6DOF head tracking in Unity to capture participant gaze and orientation in X, Y, Z coordinates throughout immersive VR sessions, enabling researchers to reconstruct full spatial trajectories for cognitive analysis.
- Collaborated with psychology PhD researchers to translate experimental design requirements into working VR software, bridging the gap between research intent and technical implementation across mixed-reality environments.

Projects

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

University Course Planning/Review Platform *(Class Best):* Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.